

OHIO STATE UNIVERSITY, OH, USA

WMO: 724288

Lat: 40.078N

Lon: 83.078W

Elev: 905

StdP: 14.22

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.5	9.1	-4.1	4.6	5.8	0.5	5.8	10.5	25.9	33.9	23.7	29.9	7.4	270	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	90.3	73.7	88.1	72.9	85.8	71.6	76.6	86.2	75.2	84.0	73.8	81.8	10.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.5	81.3	72.4	123.9	80.0	71.5	120.2	78.9	40.5	86.1	39.2	84.4	37.9	81.7	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.3	19.3	17.3	-1.8	93.0	6.7	2.3	-6.6	94.7	-10.5	96.0	-14.3	97.3	-19.2	99.0		
			DB													
			WB	-2.5	78.9	6.4	1.6	-7.2	80.1	-10.9	81.0	-14.5	81.9	-19.2	83.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	28.8	31.7	40.6	52.8	63.2	71.0	74.2	73.0	66.7	54.5	43.0	33.6
	DBStd	18.48	12.16	11.30	11.30	9.18	8.31	6.26	5.03	4.95	7.20	8.95	9.37	10.19
	HDD50	2423	664	521	338	76	6	0	0	0	53	249	516	
	HDD65	5398	1123	933	759	377	138	20	2	4	65	345	660	972
	CDD50	3466	6	8	46	158	414	631	751	712	500	192	39	9
	CDD65	967	0	0	2	10	81	201	288	251	115	19	0	0
	CDH74	8342	0	0	12	99	716	1732	2529	2067	1030	154	3	0
	CDH80	2603	0	0	1	9	176	564	860	638	323	32	0	0
Wind	WSAvg	7.9	9.6	9.3	9.4	9.3	7.7	7.2	6.3	6.0	6.1	7.4	8.3	8.8
Precipitation	PrecAvg	39.0	2.7	2.2	3.3	3.6	4.0	4.3	3.9	3.6	3.3	2.7	2.5	3.0
	PrecMax	49.1	9.2	4.2	6.4	6.6	6.8	8.5	5.8	8.0	5.6	5.7	4.4	4.6
	PrecMin	33.2	1.1	1.4	1.0	2.0	1.6	1.7	1.5	1.5	0.2	1.2	0.7	1.1
	PrecStd	4.9	2.0	0.8	1.4	1.2	1.6	1.9	1.2	1.8	1.3	1.2	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	66.9	75.5	81.4	87.4	91.3	92.8	92.0	91.4	84.1	71.5	63.7
		MCWB	56.9	58.8	62.2	64.5	70.3	72.7	76.0	74.2	71.5	69.6	61.0	58.2
	2%	DB	57.0	59.0	70.0	77.0	84.4	88.6	90.2	89.0	87.7	78.2	66.1	57.9
		MCWB	53.9	53.1	58.5	61.1	68.4	72.4	75.0	73.5	71.2	65.5	57.3	53.7
	5%	DB	52.3	54.1	63.9	72.9	81.7	86.2	87.8	86.3	83.7	73.5	62.8	53.5
		MCWB	49.1	49.2	54.7	59.2	67.4	71.1	73.7	72.4	69.7	62.9	55.7	50.2
	10%	DB	46.1	47.9	59.0	69.1	78.5	83.2	84.8	83.6	80.5	69.9	58.0	48.3
		MCWB	42.3	42.8	51.5	57.5	65.9	69.9	72.2	71.0	67.4	61.7	51.9	44.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	60.1	63.7	67.5	73.1	76.5	78.9	77.9	75.6	71.3	63.1	59.4
		MCDB	60.8	66.1	74.2	78.2	83.5	86.7	89.1	87.7	86.3	81.1	68.5	62.5
	2%	WB	54.4	54.5	59.8	63.9	71.0	74.7	77.1	76.1	73.4	68.5	59.6	54.8
		MCDB	57.1	57.5	67.7	72.7	81.4	84.4	86.9	85.2	82.4	74.9	64.3	57.4
	5%	WB	49.2	48.8	56.6	61.5	69.1	73.2	75.5	74.6	71.6	65.6	56.3	50.5
		MCDB	51.6	53.7	62.1	69.3	78.2	82.6	84.3	82.6	79.8	72.0	61.2	53.2
	10%	WB	42.6	43.1	52.7	59.2	67.4	71.6	74.1	73.1	69.9	62.3	52.8	45.3
		MCDB	46.3	47.3	58.7	66.8	75.8	80.4	81.8	80.1	76.8	68.7	57.6	48.3
Mean Daily Temperature Range	5% DB	MDBR	14.1	15.3	17.8	20.7	20.1	19.1	18.6	19.1	20.7	19.3	16.6	13.1
		MCDBR	17.9	22.7	24.1	24.9	23.3	22.0	20.5	21.1	24.3	24.0	22.4	17.7
		MCWBR	15.8	17.9	15.3	14.5	11.0	9.5	8.5	8.8	10.0	12.9	15.6	15.4
	5% WB	MCDBR	16.9	21.1	20.7	20.9	19.8	18.7	18.0	17.3	18.8	20.4	19.1	17.1
		MCWBR	16.0	18.1	14.7	13.8	10.3	9.2	8.3	8.3	8.9	11.7	15.2	16.2
Clear-Sky Solar Irradiance	taub	0.310	0.329	0.359	0.395	0.436	0.452	0.460	0.452	0.393	0.362	0.327	0.305	
	taud	2.412	2.337	2.328	2.251	2.176	2.176	2.176	2.165	2.168	2.312	2.388	2.447	2.450
	Ebn at Noon	270	280	283	279	268	263	260	259	268	265	261	263	
	Edh at Noon	26	32	36	42	47	47	47	45	37	30	25	23	
All-Sky Solar Radiation	RadAvg	517	792	1107	1496	1741	1910	1869	1721	1400	966	649	439	
	RadStd	49	99	102	130	144	134	108	86	125	85	85	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+122

Nomenclature: See separate page